

MFE 230G – Final Project Assignment

Deadline: September 10th, 2025

Goal

Investigate and evaluate an active investment strategy of your choice, leveraging industry/factor returns, emissions intensity data, or broker/alpha signals. The key requirement: integrate ChatGPT as your AI research assistant for ideation, implementation support, and robustness checks.

Deliverables

Your final report (individual or group of up to 5) must follow the Final Report Template:

1. Executive Summary (1 paragraph)
 - Clearly state your proposed strategy.
 - If results do not support alpha generation, explicitly conclude “Do not implement” with justification.
2. What Did You Try? (2–5 pages)
 - a. Ideas: Origin of your strategy (emissions tilts, momentum, IC-based alphas, macro drivers, etc.)
 - b. Data: Describe sources. Suggested data:
 - i. Ken French Data Library: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
 - ii. BEA Input-Output Accounts: <https://www.bea.gov/industry/input-output-accounts-data>
 - iii. FRED: <https://fred.stlouisfed.org/>
 - iv. BEA Economic Accounts: <https://www.bea.gov/data/economic-accounts>
 - v. BLS Employment & Output: <https://www.bls.gov/emp/tables/industry-employment-and-output.htm>
 - c. Implementation: Risk modeling, turnover and costs, and ChatGPT prompts (≥3 high-quality prompts).
3. What Did You Learn? (1–4 pages):
 - a. Style/risk factor exposures
 - b. Time patterns of performance (full sample, post-2010, recent 12–18 months).
 - c. Robustness and limits of your strategy

- d. ChatGPT Evaluation: Critically assess usefulness, flaws, and adaptations.
4. Appendices:
 1. Include ChatGPT transcripts, tables/plots, and code snippets.

Connection to Homeworks

- HW01 (Emissions Exercise): You practiced using ChatGPT to generate two candidate strategy ideas based on high vs. low emitters and factor exposures.
- HW02 (Alpha Optimization): You extended the practice by using ChatGPT to critique alpha assumptions and robustness testing. Apply the same critical mindset when designing your project strategy.

Evaluation Criteria

- Strength & clarity of thesis (40%) – Is the strategy well-motivated, consistent with the data, and clearly explained?
- Execution (40%) – Risk control, horizon analysis, robustness checks. Did you critically evaluate ChatGPT's role?
- Originality (20%) – Novel data use, prompt design, or unique insights. Bonus credit for creative but well-structured ChatGPT use.

ChatGPT Requirement

Each project must:

1. Use at least three substantial ChatGPT interactions (idea generation, coding support, robustness design).
2. Document each interaction: Prompt(s), summarized output, and your critical evaluation.
3. Reflect on ChatGPT's strengths/limits as an investment research assistant.

Suggested Drivers and Reading

Industry Momentum:

- Zarattini & Antonacci, 'A Century of Profitable Industry Trends' (SSRN):
- https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4857230

Industry Links:

- MSCI Blog: Understanding the Industry:
- <https://www.msci.com/www/blog-posts/understanding-the-industry/02044436215>

Strategy Examples:

- S&P Global: Sector Rotation Strategy Index:
- <https://www.spglobal.com/spdji/en/documents/education/brochure-sector-rotation-strategy-index.pdf>
- Industry Predictability Research (PM-Research):
- <https://www.pm-research.com/content/iijpormgmt/44/1/30>

Machine Learning Approaches:

- NBER Working Paper w30217: ML in Asset Pricing:
- https://www.nber.org/system/files/working_papers/w30217/w30217.pdf